Speech Disfluency as a Thought–Language Marker of Early Psychosis

Spontaneous speech offers a direct window into the dynamics of language and thought. Unfilled pauses and fillers (e.g., “uhm”) reflect the moment-by-moment unfolding of cognitive processes (1; 2). Pauses between 250–3,000 ms are typical (3; 4), and their syntactic position reveals where thinking and linguistic formulation become effortful. Utterance-initial pauses signal planning (5; 6), while within-utterance pauses reflect lexical retrieval or syntactic effort (7; 8). Fillers relate to speech monitoring and social signalling (6; 9). Since psychosis involves disruptions in thought organization (2; 10), disfluency patterns offer a sensitive marker of cognitive-linguistic dysfunction (2; 11). However, most evidence comes from chronic patients, with limited differentiation of pause types and insufficient grounding in models of language-thought interaction. Therefore, we examined whether specific syntactically defined disfluencies reflect altered thought and language processes in **first-episode psychosis (FEP) and ultra-high-risk (UHR) individuals**, and whether UHR individuals exhibit early signs of the FEP profile.

Participants: The sample included 50 FEP participants, 50 UHR, and 34 neurotypical controls (NCs). FEP participants had a history of a first psychotic episode within the last 12 months. UHR participants had brief intermittent psychotic symptom syndrome, attenuated positive symptom syndrome, and genetic risk. **Task:** Participants produced 1-minute descriptions of four images from the Thought and Language Index (12), a tool explicitly designed to probe thought–language.

Automatic feature extraction: We classify pauses following (2; 9), discarding pauses that are either too short (<250ms, as they are considered phonatory gaps linked to the respiratory cycle) or too long (>3s, as they signal a participant was no longer engaged with the given task). Recordings were processed using Whisper (13) and CrisperWhisper (14) to auto-transcribe, obtain word-level timing information, and speech disfluencies. Annotation followed prior protocols (2). Pauses were tagged by syntactic position: utterance-initial pause (e.g. [PAUSE] *He phoned his friend)*, within-utterance pause (e.g., *He phoned* [PAUSE] *his friend.)*. Duration of pauses was also indexed. Additionally, filled pauses were tagged (e.g., [FILLER: er] He phoned his friend.).

Predictions: We predicted the FEP and UHR groups would exhibit greater overall dysfluency—particularly at within utterance. The FEP and UHR groups would produce fewer fillers.

Results: Our Poisson regression model showed that fillers were significantly reduced in both FEP ($\beta = -0.60$, $p < .001$) and UHR ($\beta = -0.35$, $p < .001$) compared to NCs (Fig 1). Because fillers often function as metacognitive markers of monitoring and turn-holding, their reduction may reflect diminished self-regulation in speech production, consistent with weakened higher-order thought organization. In contrast, total unfilled pauses were substantially higher in FEP ($\beta = 0.50$, $p < .001$) and UHR ($\beta = 0.44$, $p < .001$). Despite producing more pauses overall, utterance-initial pauses were reduced in FEP ($\beta = -0.17$, $p < .001$), suggesting impaired speech-planning processes, and potentially reflecting disrupted initiation or reduced anticipatory structuring of thought before articulation. Within-utterance pauses were markedly increased in both FEP ($\beta = 0.43$, $p < .001$) and UHR ($\beta = 0.38$, $p < .001$). These pauses are tightly linked to lexical retrieval, and syntactic integration. Their elevation suggests difficulty coordinating linguistic units during thought formulation: a pattern consistent with increased processing demands or disruptions in cognitive control mechanisms supporting language. Repetitions were higher in FEP ($\beta = 1.68$, $p < .001$), indicating possible breakdowns in the working memory, and self-monitoring: cognitive systems essential for maintaining coherent thought-speech mappings. Overall, these findings reveal distinct dysfluency profiles in FEP and UHR relative to NCs. Reduced fillers, increased pauses (particularly within-utterance), and elevated repetitions align with theoretical accounts proposing attenuated self-monitoring, altered predictive control, and compromised linguistic planning (10). This pattern supports the conclusion that the social-signalling and interactive functions of speech are diminished in early psychosis and ultra-high risk (2; 10). Consequently, we propose that such dysfluencies represent sensitive behavioural indicators of disruptions in thought, language, and cognition, offering non-invasive markers of early psychotic vulnerability.

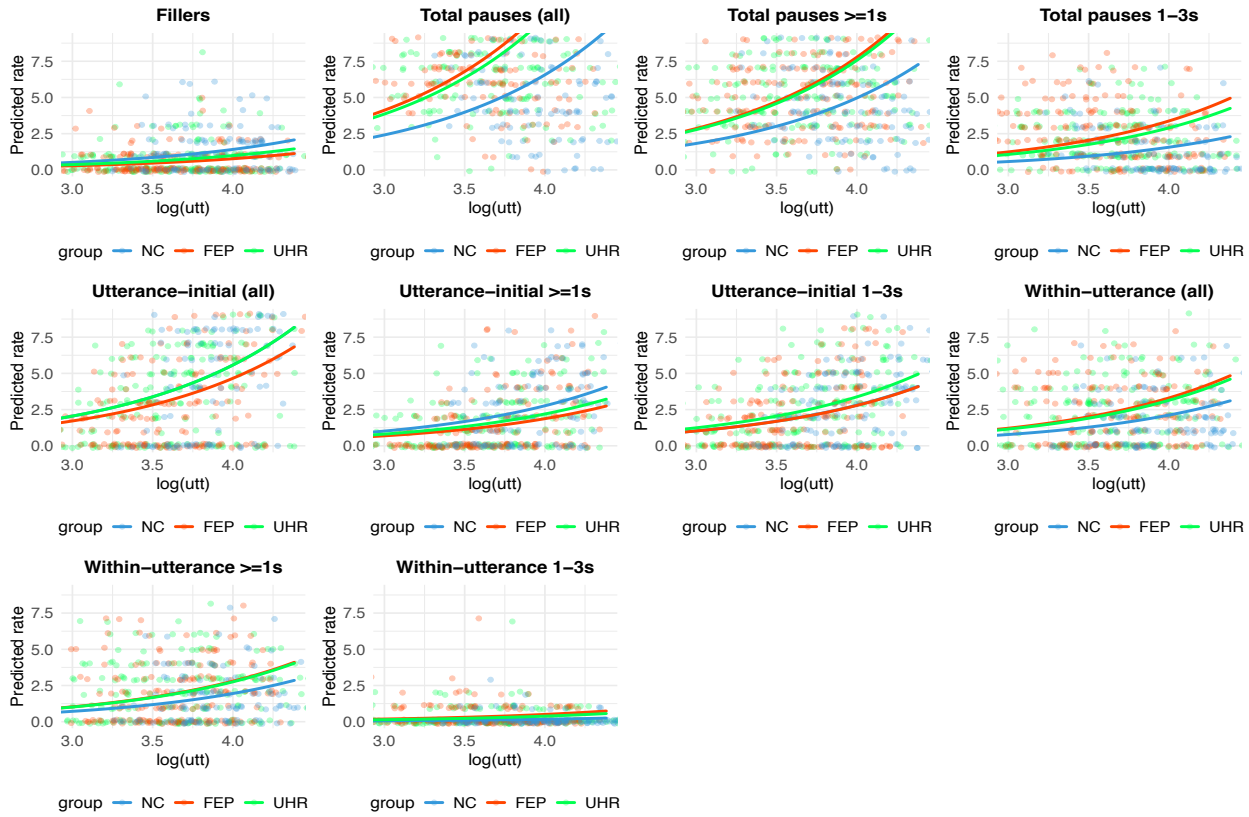

Figure 1. Predicted rates of speech dysfluencies by log (utterance) and group. Age, education and gender were co-variants in the Poisson regression model. Note: In utterance-initial (all) and utterance-initial >=1s, NC's regression line overlaps with UHR's line and FEP's line respectively.

1. Bello-Lepe, S., Mahmood, S., Varley, R., & Zimmerer, V. (2024). *Cortex*, *178*, 287–298.
2. Çokal, D., Zimmerer, V., Turkington, D., Ferrier, N., Varley, R., Watson, S., & Hinzen, W. (2019). *PLOS ONE*, *14*(5), e0217404.
3. Goldman-Eisler, F. (1968). *Psycholinguistics; experiments in spontaneous speech. London.*
4. Schegloff, E. A., Jefferson, G., & Sacks, H. (1977). *Language*, *53*(2), 361–382.
5. Butterworth, B. (1979). *Brain and Language*, *8*(2), 133–161.
6. Levelt, W. J. M. (1989). *Speaking: From intention to articulation*. The MIT Press.
7. Hartsuiker, R. J., & Notebaert, L. (2010). *Experimental Psychology*, *57*(3), 169–177.
8. Kircher, T. T. J., Brammer, M. J., Levelt, W., Bartels, M., & McGuire, P. K. (2004). *NeuroImage*, *21*(1), 84–90.
9. Howes, C., Lavelle, M., Healey, P. G. T., Hough, J., & McCabe, R. (2017). *CogSci.*
10. Andreasen, N. C. (1986). *Schizophrenia Bulletin*, *12*(3), 473–482.
11. Parola, A., Simonsen, A., Bliksted, V., & Fusaroli, R. (2020). *Schizophrenia Research*, *216*, 24–40.
12. Liddle, P. F., Ngan, E. T. C., Caissie, S. L., Anderson, C. M., Bates, A. T., Quested, D. J., White, R., & Weg, R. (2002). *The Journal of Mental Science*, *181*, 326–330.
13. Radford, A., Kim, J. W., Xu, T., Brockman, G., McLeavey, C., & Sutskever, I. (2022). *Robust Speech Recognition via Large-Scale Weak Supervision*.
14. Zusag, M., Wagner, L., & Thallinger, B. (2024). *CrisperWhisper: Accurate Timestamps on Verbatim Speech Transcriptions*. 1265–1269.